Solve and Graph Inequalities

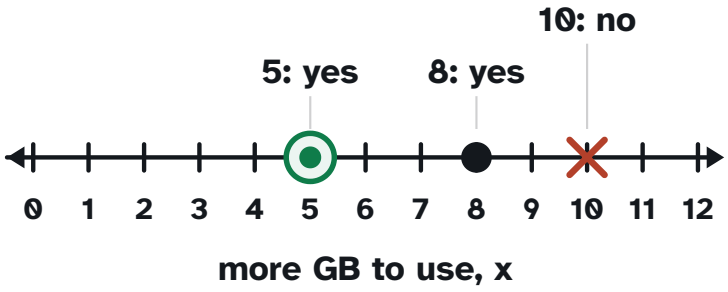
Solve an inequality first, then shade the graph from the solved symbol.

Name _____ **Date** _____

Tonight's spark

A family phone plan allows at most 20 GB of data each month. The bill shows 12 GB already used. Can the family use 10 more GB?

Guess first: can they use 10 more GB, yes or no?



Our class model: shade from the solved symbol

Try it

Work it out on paper. Write one answer on each line.

1 **Solve: $x - 4 > 2$**

A. $x > 2$ **B.** $x > -2$ **C.** $x > 6$

My answer _____

2 **Solve: $5x \leq 30$**

A. $x \leq 6$ **B.** $x \leq 25$ **C.** $x \leq 150$ **D.** $x \geq 6$

My answer _____

3 **After solving you get $n \geq 3$. Which graph shows it?**

A. Closed circle at 3, shade left **B.** Open circle at 3, shade right

C. Open circle at 3, shade left **D.** Closed circle at 3, shade right

My answer _____

4 **Solve: k ÷ 2 > 5**

A. k > 7 **B.** k > 10 **C.** k > 2.5

My answer _____

Talk about it

Which number goes at the circle, the one you started with or the solved one? Why?

Home mission

Look at a phone bill or gift card. Pretend a plan allows at most 20 GB and 12 are used. Write x + 12 ≤ 20 and solve it. Test one number that works and one that does not.

If we get stuck, ask

- Solve it first. What does the symbol say NOW?
- Does that symbol point toward bigger or smaller numbers?
- Test a shaded value in the original. Is it true?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/8-6-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it